Ayush Dhakad

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.42
Kiddy's corner higher secondary school	Class XII (Central Board of Secondary Education (CBSE))	2023	84.60%
st. Paul's School	Class X (Central Board of Secondary Education (CBSE))	2021	87.20%

WORK EXPERIENCE

Machine Learning Research Intern | DENSO

Computer Vision & Topology Modeling

- Built a 2-stage CV pipeline mapping **2D RGB** images into **structured plant topology graphs** for harvesting robot.
- Applied the **Separating Axis Theorem (SAT)** for collision detection to construct edges between **OBBs**.
- Implemented the **Hungarian Algorithm** for one-to-one component pairing across reconstructed plant topologies.

Synthetic Data & Generative AI

- Built a synthetic data engine using Blender, **Stable Diffusion**, and **ControlNet** to generate 40,000+ annotated images.
- Generated conditional inputs (Canny edge maps and segmentation masks) via procedural Blender workflows.
- Fine-tuned **ControlNet** via **LoRA** adaptation to bridge the sim-to-real domain gap and generate photo-realistic images.

Object Detection & Evaluation

- Benchmarked sim-to-real transfer by training YOLO-OBB on synthetic images and validating on real images.
- Achieved **0.70 F1-score**, with **93%** large-object and **80%** small-object precision across greenhouse images.

PROJECTS

Dynamic Rerouting: Real-Time Incremental Pathfinding Engine

Python, NetworkX, D Lite, Dijkstra, Monte Carlo Simulation*

- Implemented **D* Lite algorithm** from scratch in Python to enable incremental real-time navigation across large-scale grid graphs (Tested on 450,000+ nodes and 820,000+ weighted edges).
- Built a **Monte Carlo disruption engine** simulating real-time road closures to evaluate dynamic path recovery.
- Benchmarked incremental replanning against Dijkstra algorithm, achieving a **98% reduction in node expansions** (25k vs 2M) and a **2.0x runtime speedup** across 50 consecutive dynamic disruptions.

Codebase Graph RAG System

Python, Tree-sitter, Neo4j, Gemini API

- Built a **Graph RAG** system to parse codebases into **AST** graphs linking functions, classes, imports, and calls.
- Implemented graph-based retrieval to return relevant, dependency-connected functions instead of the full repository.
- Reduced live Gemini token usage by **94%** (9,425 to 563 tokens) while retaining correct answers.

Offline Multi-Hop Bluetooth Messenger

Python, BLE, AsyncIO, Cryptography

- Built **offline P2P messenger** enabling multi-hop Bluetooth message forwarding without internet.
- **Secured communications** using end-to-end encryption, message auth, TTL routing, and duplicate suppression.
- Dynamic relaying via asynchronous device discovery and store-and-forward routing across shifting nodes.

Structural Topology Optimization Using U-net

Course Project under Prof. Surendra Nath Somala

- Designed a 5-channel U-Net to predict structural layouts from boundary conditions.
- Added volume and binarization penalties to the loss to encourage material limits and solid predictions.
- Evaluated the model on 500 unseen geometries, reporting inference under 30 ms.

SKILLS

Programming Languages: C++, Python, C, SQL

ML/DL: NumPy, Pandas, Matplotlib, Scikit-learn, Tensorflow, Keras, Deep Neural Networks

CS Fundamentals: DSA(**500+** Questions solved on Leetcode and Codeforces), OOPs, DBMS, Computer Networks

RELEVANT COURSES

Programming: C++, Python, Data-Structure and Algorithm

Machine Learning: Machine Learning And Scientific Computing For Civil Engineers, Supervised/Unsupervised Learning, Deep Learning, Basic Robotics

Mathematics: Calculus 1, Calculus 2, Elementary Linear Algebra, Differential Equations, Transform Techniques, Probability, Statistics, Complex Variables

Languages: Spoken Japanese Basics

POSITIONS OF RESPONSIBILITY

- Internship and Placement Coordinator IIT Hyderabad
- Assistant Manager, E-cell IIT Hyderabad
- Multimedia Team Coordinator, Elan & nVision (College Fest)

EXTRACURRICULAR

- Participated in a Hackathon organized by NPCI
- Developed an AI Hangman solver at the Meltwater hackathon
- Authored a technical research paper summarizing internship findings.
- Active member of NSO (National Sports Organisation), Athletics
- Pursue photography and videography as creative interests